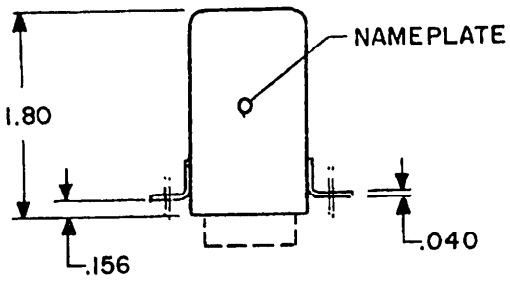
MILITARY SPECIFICATION SHEET

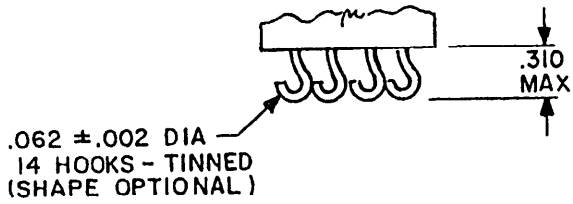
RELAYS, TIME DELAY ON RELEASE, SEPARATE POSITIVE CONTROL TYPE IIA, CLASS B, 10 AMP, 2PDT, HERMETICALLY SEALED

INACTIVE FOR NEW DESIGN AFTER 7 September 1984
USE MIL-R-83726/29(USAF)

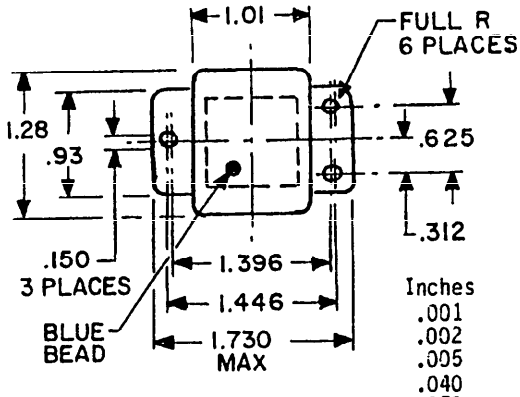
This specification is approved for use by all Departments and Agencies of the Department of Defense.

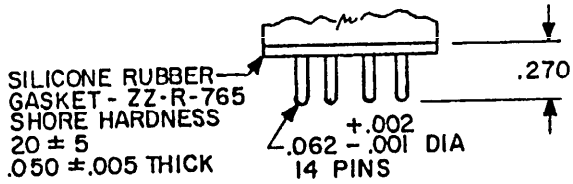
The requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-R-83726.





SOLDER HOOK TERMINAL

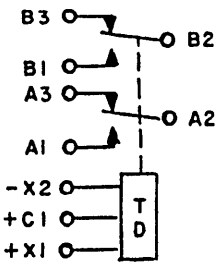




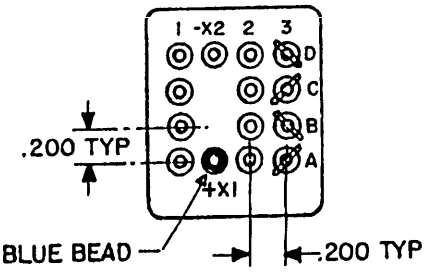
GOLD PLATE PER
MIL-G-45204, TYPE II, CLASS I, UNDER PLATING
NICKEL 50 to 150 MICROINCHES THICK.

PLUG-IN TERMINAL

Inches	mm
.001	0.03
.002	0.05
.005	0.13
.040	1.02
.050	1.27
.062	1.57
.150	3.81
.156	3.96
.200	5.08
.270	6.86
.310	7.87
.312	7.92
.625	15.88
.93	23.62
1.01	25.65
1.28	32.51
1.396	35.46
1.446	36.73
1.730	43.94
1.80	45.72



CIRCUIT DIAGRAM



TERMINAL LAYOUTS

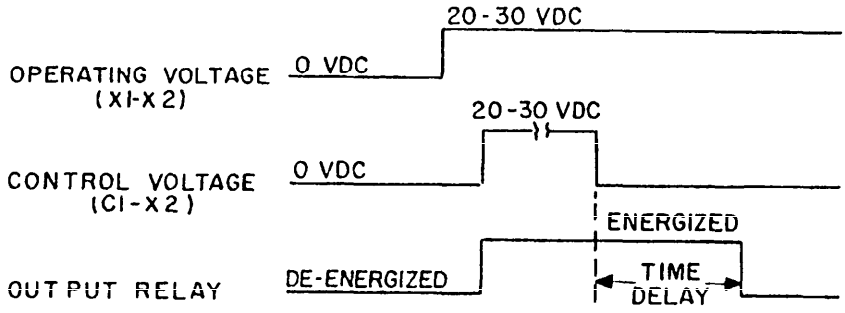
HOOK TERMINALS TO BE SET AT ANGLES SHOWN IN ROW 3. EACH LETTERED ROW TO FACE SAME DIRECTION
HOOK TERMINAL SPACING TOLERANCE ±.03 (0.76 mm)

Ⓐ FIGURE 1. Outline dimensions and configuration.

Ⓐ denotes changes

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TIMING DIAGRAM

NOTES:
1. Dimensions in inches.
2. Unless otherwise specified, tolerances are ±.03 (0.8 mm) on two place decimals and ±.010 (0.25 mm) on three place decimals.
3. Metric equivalents are given for general information only.
4. Terminal numbers need not appear on the relay header provided there is affixed to the relay a permanent legible circuit diagram that identifies each terminal location specified. C2, C3, and D terminals are not used in internal circuitry and may be omitted from terminal layout.
5. During timing cycle, relay will recycle if control voltage is reapplied and removed from terminal C1.

Ⓐ FIGURE 1. Outline dimensions and configuration - Continued.

Ⓐ TABLE I. Dash number and characteristics.

M83725/2		Time	M83726/2		Time	M83725/2		Time
Solder hook	Plug-in	delay seconds ±10%	Solder hook	Plug-in	delay seconds ±10%	Solder hook	Plug-in	delay seconds ±10%
-001	-101	.100	-020	-120	25.0	-038	-138	130.0
-002	-102	.500	-021	-121	30.0	-039	-139	140.0
-003	-103	.750	-022	-122	35.0	-040	-140	150.0
-004	-104	1.0	-023	-123	40.0	-041	-141	160.0
-005	-105	2.0	-024	-124	45.0	-042	-142	170.0
-006	-106	3.0	-025	-125	50.0	-043	-143	180.0
-007	-107	4.0	-026	-126	55.0	-044	-144	190.0
-008	-108	5.0	-027	-127	60.0	-045	-145	200.0
-009	-109	6.0	-028	-128	65.0	-046	-146	210.0
-010	-110	7.0	-029	-129	70.0	-047	-147	220.0
-011	-111	8.0	-030	-130	75.0	-048	-148	230.0
-012	-112	9.0	-031	-131	80.0	-049	-149	240.0
-013	-113	10.0	-032	-132	85.0	-050	-150	250.0
-014	-114	12.0	-033	-133	90.0	-051	-151	260.0
-015	-115	14.0	-034	-134	95.0	-052	-152	270.0
-016	-116	16.0	-035	-135	100.0	-053	-153	280.0
-017	-117	18.0	-036	-136	110.0	-054	-154	290.0
-018	-118	20.0	-037	-137	120.0	-055	-155	300.0
-019	-119	22.0						

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Transient voltage limits (input)

REQUIREMENTS:

Contact data:

Configuration: 2 PDT

Life load ratings (relay case grounded): See table II.

Ⓐ TABLE II. Life/load rating (relay case grounded).

Type of load	Life (Cycles)	Amperes	
		23 V dc	115/200 volts 1 and 3 phase 400 Hz
Resistive	100,000	10	10
Inductive	20,000	8	8
Motor	100,000	4	4
Lamp	100,000	2	2
Reduced current	400,000	2.5	2.5

Contact voltage drop:

Initial: 0.150 volt.

After life tests: 0.175 volt.

Minimum current: Applicable in accordance with MIL-R-6106 specification.

Contact bounce: 1 millisecond maximum.

Overload:

dc: 40 amperes.

ac: 60 amperes.

Rupture:

dc: 30 amperes.

ac: 30 amperes.

Input data:

Duty rating: Continuous.

Maximum voltage:	Over temperature range	30 V dc.
Nominal voltage:		28 V dc.
Minimum voltage:		20 V dc.

Minimum voltage high temperature test: 21 V dc.

Minimum voltage continuous current test: 23.5 V dc.

Maximum current at 25°C: 0.150 ampere (X1).

Maximum control current at 25°C: 0.015 ampere (C1)

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Operate time at 25°C 28 V dc: 10 milliseconds maximum.

Release time: See table I.

Recycle time: 50 milliseconds maximum.

Electrical data:

Insulation resistance at 500 V dc 1/:

Initial: 1,000 megohms minimum.

After life or environmental tests: 500 megohms minimum.

Dielectric strength (sea level):

	Initial	After life tests
Input (X1, C1, X2) 1/:	1,000 V rms	1,000 V rms
All other points:	1,250 V rms	1,000 V rms

Dielectric strength (altitude): 80,000 feet 1/ and 2/.

Input (X1, C1, X2): 350 V rms.

All other points: 350 V rms.

Transients:

Transient voltage limits (input):

Surge positive MIL-STD-704: 80 V dc maximum 3/.

Power loss: 500 microseconds maximum.

Spike: MIL-STD-704.

Self generated: ±50 V maximum.

Susceptability: ±600 V maximum 4/.

RFI: MIL-STD-461, class 1D.

Environmental data:

Temperature range: -65°C to +125°C.

Maximum altitude rating: 80,000 feet.

Shock G-level: 100 G.

Duration: 6 microseconds.

Maximum duration: 10 microseconds, monitor in accordance with method 310 of MIL-STD-202.

1/ Input and control terminals X1, C1, and X2 must be connected together during this test.

2/ Dielectric rating may be improved by suitable insulation of terminals and wiring after installation, .500 V rms at 80,000 feet when properly mounted in mating socket.

3/ In accordance with MIL-STD-704A, figure 9, limit 1, duty cycle 2 percent.

4/ In accordance with MIL-STD-704A, figure 17, 0 to 500 microseconds, duty cycle 2 percent.

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Vibration (sinusoidal):

G-level: 20 G.

Frequency range: 10 to 3,000 Hz

Vibration (random):

Applicable specification: MIL-STD-810.

Power spectral density: 0.4 G^2/Hz.

Frequency range: 50 to 2,000 Hz.

Acceleration: 20 G.

Ⓐ Seal: Not applicable

Physical data:

Weight: 0.32 pound maximum.

Dimensions and configuration: See figure 1.

Ⓐ Mating socket: MIL-S-12883/40-01 and 07 (Caution: Consideration should be given to ambient temperature and current requirements when using wire barrels size 20.)

Part number: Consists of the basic number of this specification sheet and a dash number.

Example:

M83726/2 - 001

Basic document number ——— Dash number from table I.

Ⓐ Quality conformance inspection:

Periodic inspection: Performance of group B and C inspections are not applicable.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5945-0756-2)

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